

Detecting accidental awareness during general anaesthesia with neuromuscular blockade

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Introduction

Consciousness

- ▶ **Consciousness**, our subjective experience, can be defined as “what abandons us every night and returns the next morning when we wake up or when we dream”¹
- ▶ Encompassing all subjective experience this can be pure darkness, an exciting movie, or intraoperative pain
- ▶ Patients and anaesthesiologists assume that general anaesthesia implies an experience of unconsciousness similar to dreamless sleep

¹Tononi, G. (2004). An information integration theory of consciousness. BMC neuroscience, 5, 1-22.

Connectedness

- ▶ Environmental **connectedness** refers to the connection between consciousness and the external world which allows us to experience external stimuli
- ▶ Importantly consciousness can be disconnected (e.g. during dreams) or connected (e.g. normal wakefulness)

Responsiveness

- ▶ Behavioural interaction with the “outside world” is considered **responsiveness**
- ▶ This can be further subdivided into spontaneous (e.g. self initiated movements) or goal-directed responsiveness (e.g. command following)

Examples

- ▶ Wakefulness: conscious, connected and responsive
- ▶ NREM sleep: unconscious, unconnected, unresponsive
- ▶ REM sleep: conscious (during dreams), unconnected, unresponsive

Relevance for anaesthesia monitoring

- ▶ In normal sleep-wake states connection to the environment quickly leads to responsiveness
- ▶ During anaesthesia the coupling between responsiveness and connectedness is sometimes lost²
- ▶ This is important because the primary aim of anaesthesia is to *suppress the experience of surgery* either via unconsciousness or disconnection (dreaming would not be a problem)
- ▶ The other aims are nociceptive blockade and immobility
- ▶ It is also important to note that the aim is not to merely induce amnesia of the events (even though that's not undebated)
- ▶ **Accidental awareness during general anaesthesia (AAGA)** is a state of environmentally connected consciousness

²Sanders, R. D., Tononi, G., Laureys, S., Sleigh, J. W., & Warner, D. S. (2012). Unresponsiveness $\neq$ unconsciousness. The Journal of the American Society of Anesthesiologists, 116(4), 946-959.

Awareness during anaesthesia

- ▶ In isolated forearm technique (IFT) a cuff is placed on the forearm to prevent **neuromuscular blockade** of the arm allowing the patient to communicate via movements (“If you can hear me squeeze my hand.”, an example of **goal-directed responsiveness**)
- ▶ Only 0.2% of patients recall awareness during anaesthesia³ in contrast to 37%* of patients who respond to commands using the IFT⁴
- ▶ Fortunately pain was rarely reported
- ▶ ***The problem: current anaesthesia monitors (such as the bispectral index) cannot detect responders to the IFT⁵ and as a consequence may be unable to detect AAGA***

³Sandin, R. H., Enlund, G., Samuelsson, P., & Lennmarken, C. (2000). Awareness during anaesthesia: a prospective case study. *The Lancet*, 355(9205), 707-711.

⁴Sanders, R. D., Tononi, G., Laureys, S., Sleight, J. W., & Warner, D. S. (2012). Unresponsiveness $\neq$ unconsciousness. *The Journal of the American Society of Anesthesiologists*, 116(4), 946-959.

⁵Schneider, G., Gelb, A. W., Schmeller, B., Tschakert, R., & Kochs, E. (2003). Detection of awareness in surgical patients with EEG-based indices—bispectral index and patient state index. *British journal of anaesthesia*, 91(3), 329-335.

Monitoring with the bispectral index⁶



	IFT response	No IFT Response
BIS response	47	220
No BIS response	33	1273

⁶Russell, I. F. (2013). The ability of bispectral index to detect intra-operative wakefulness during total intravenous anaesthesia compared with the isolated forearm technique. *Anaesthesia*, 68(5), 502–511.
<https://doi.org/10.1111/anae.12177>

Amnesia

- ▶ Dosages of general anaesthetics needed to induce unconsciousness cause amnesia as a secondary effect
- ▶ Memory is not essential for experience
- ▶ Patients with memory formation deficits (see case of H.M.) still have conscious experiences
- ▶ I assume most patients would chose to not experience an event instead of experiencing and then not remembering it
- ▶ Thus the **lack of connected consciousness** should be a goal for the delivery of general anaesthesia

Dosage

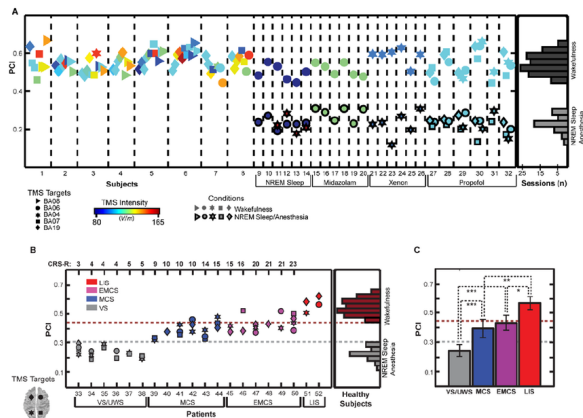
- ▶ Why not use high dosages for all patients?
- ▶ Anaesthesia has cardiovascular side effects (decreased cardiac output, decreased blood pressure, altered vascular tone) and may not be safely achievable
- ▶ Critical organs such as brain, heart and kidneys perform have the ability to regulate blood flow to maintain a constant supply (autoregulation) which may be impacted by anaesthesia
- ▶ There are some indications that depth of anaesthesia may be correlated with morbidity⁷
- ▶ Thus a second goal should be delivering general anaesthesia with the **minimal dosage** needed for disconnected consciousness

⁷Monk, T. G., Saini, V., Weldon, B. C., & Sigl, J. C. (2005). Anesthetic management and one-year mortality after noncardiac surgery. *Anesthesia & Analgesia*, 100(1), 4-10.

Alternatives to BIS

Perturbational methods⁸

- ▶ One of the most reliable methods to detect consciousness uses transcranial magnetic stimulation (TMS) to perturb the cortex
- ▶ EEG is used to record the response and an index of complexity (the perturbational complexity index; PCI) is computed from this response
- ▶ The disadvantage of this method is the time needed for application (~200 pulses delivered with ~0.5 Hz requires 6-8 minutes) and the presence of the device itself



⁸Casali, A. G., Gosseries, O., Rosanova, M., Boly, M., Sarasso, S., Casali, K. R., ... & Massimini, M. (2013). A theoretically based index of consciousness independent of sensory processing and behavior. *Science translational medicine*, 5(198), 198ra105-198ra105.

Methods derived from the spontaneous EEG: Signal diversity⁹

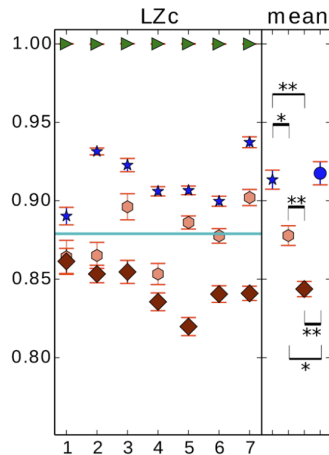
(a) a_i :  $= \text{abs}(\text{hilbert}(x_i))$

$$b_i(t) = \begin{cases} 1 & \text{if } a_i(t) > \text{mean}(a_i) \\ 0 & \text{if } a_i(t) \leq \text{mean}(a_i) \end{cases}$$

(b) b_i : ...01010010000...

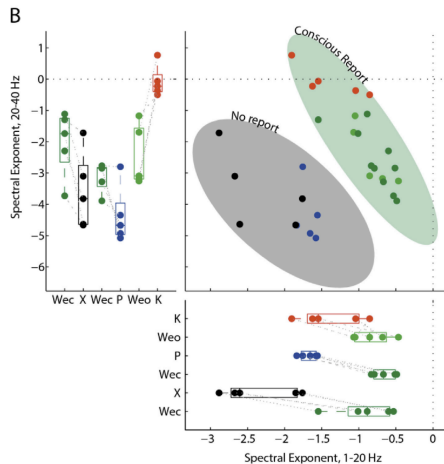
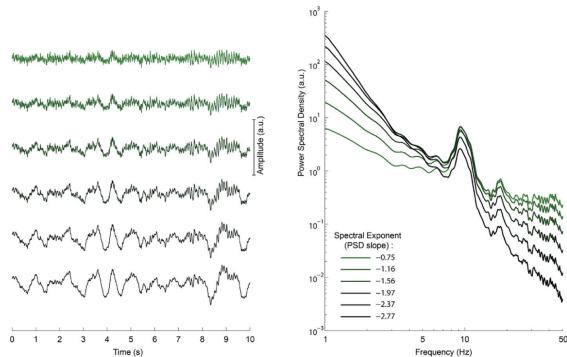
(c) $\begin{pmatrix} 000010000000011000101101... \\ 010100100001100101010111... \\ \vdots \\ 1011100101100100000001110... \end{pmatrix}$
 $n \times m$
 binary matrix $[b_i(t)]$
 of n time series and
 m observations
 column-by-column
 concatenated
 into one sequence s
 $s: b_1(1), b_2(1), \dots, b_n(1), b_1(2), b_2(2), \dots, b_n(2), \dots, b_n(m)$

(d) $\{110,001,\dots\}$ = dictionary of binary "words" in s
 $\text{LZc} \propto |\{110,001,\dots\}|$



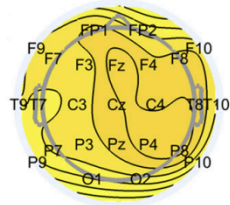
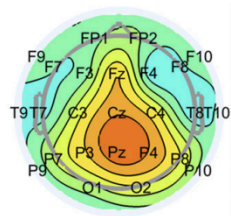
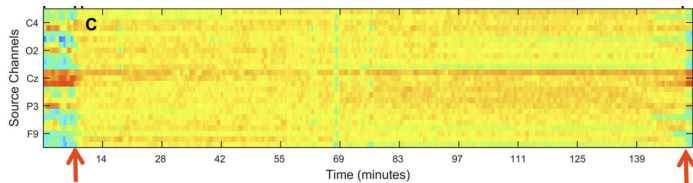
⁹Schartner, M., Seth, A., Noirhomme, Q., Boly, M., Bruno, M. A., Laureys, S., & Barrett, A. (2015). Complexity of multi-dimensional spontaneous EEG decreases during propofol induced general anaesthesia. PloS one, 10(8), e0133532.

Methods derived from the spontaneous EEG: Slope¹⁰



¹⁰Colombo, M. A., Napolitani, M., Boly, M., Gosseries, O., Casarotto, S., Rosanova, M., ... & Sarasso, S. (2019). The spectral exponent of the resting EEG indexes the presence of consciousness during unresponsiveness induced by propofol, xenon, and ketamine. *NeuroImage*, 189, 631-644.

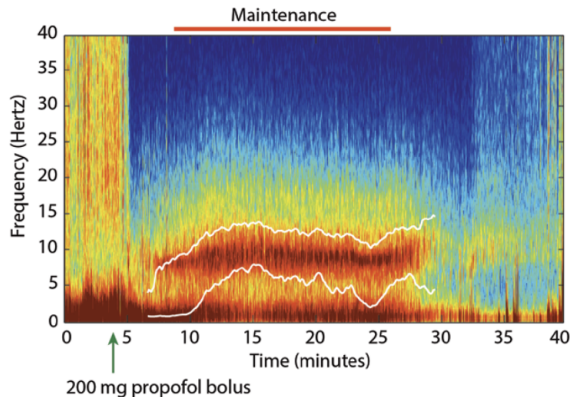
Methods derived from the spontaneous EEG: Connectivity¹¹



¹¹Juel, B. E., Romundstad, L., Kolstad, F., Storm, J. F., & Larsson, P. G. (2018). Distinguishing anesthetized from awake state in patients: a new approach using one second segments of raw EEG. *Frontiers in Human Neuroscience*, 12, 40.

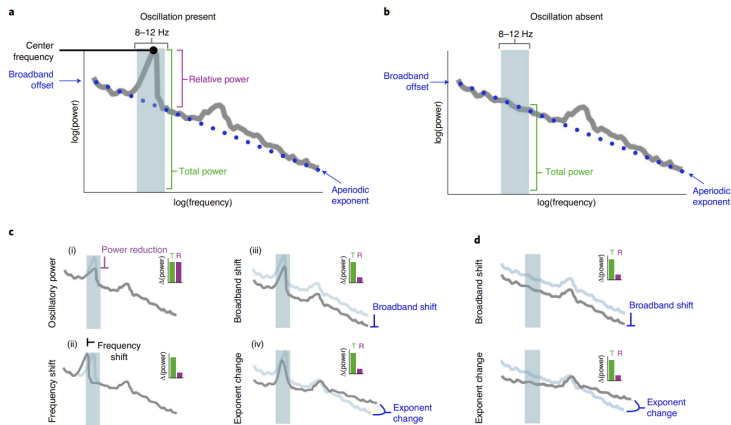
Methods derived from the spontaneous EEG: PSD¹²

A Propofol-Induced Unconsciousness: Slow and Alpha Oscillations



¹²Purdon, P. L., Sampson, A., Pavone, K. J., & Brown, E. N. (2015). Clinical electroencephalography for anesthesiologists: part I: background and basic signatures. *Anesthesiology*, 123(4), 937-960.

Periodic and Aperiodic Components of the PSD with FOOOF¹³



¹³Donoghue T, Haller M, Peterson EJ, Varma P, Sebastian P, Gao R, Noto T, Lara AH, Wallis JD, Knight RT, Shestyuk A, & Voytek B (2020). Parameterizing neural power spectra into periodic and aperiodic components. *Nature Neuroscience*, 23, 1655-1665.

So there are lot's of methods that work?

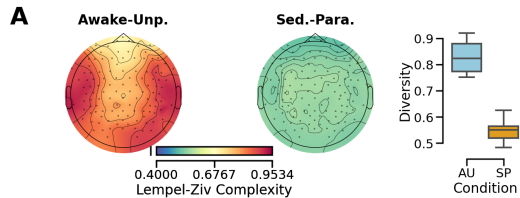
Data

- ▶ 115 EEG channels recorded at 5000 Hz (inside a Faraday cage)
- ▶ Six healthy volunteers (one female)
- ▶ Sedative agents: Fentanyl (100 μg), Propofol (Bolus 37-87 mg; target 2-4 mcg ml^{-1}) via IV bolus (10-26 s duration)
- ▶ Eyes closed entire experiment
- ▶ Downsampling to 250 Hz, re-referencing to REST (Yao, 2001), 1 Hz highpass
- ▶ Extracted 120 s of continuous EEG and split into 5 s epochs with 50% overlap (47 epochs per participant per condition)
- ▶ The 120 s of the 'sedation' condition were extracted 240 s after the drugs were administered
- ▶ Trained linear kernel SVM on 'Awake-Unparalysed' vs. 'Sedated-Paralysed'
- ▶ Evaluation via leave-one-participant-out (transfer learning)

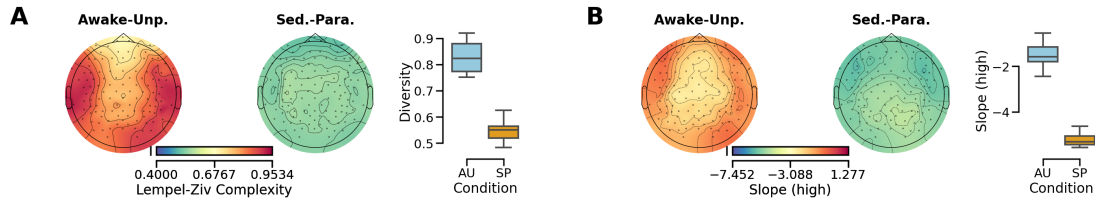
Features

- ▶ Diversity (LZc, channel-wise, 1-45 Hz; see *Schartner et al., 2015*)
- ▶ Slope (exponent computed with method from *Colombo et al., 2019* for canonical bands + “low” and “high”)
- ▶ Outflow (DTF connectivity metric, 8-12 Hz, 26 channels; see *Juel et al., 2018*)
- ▶ PSD (periodic components computed with FOOOF, canonical bands + 1-45 Hz; see *Donoghue et al., 2020*)

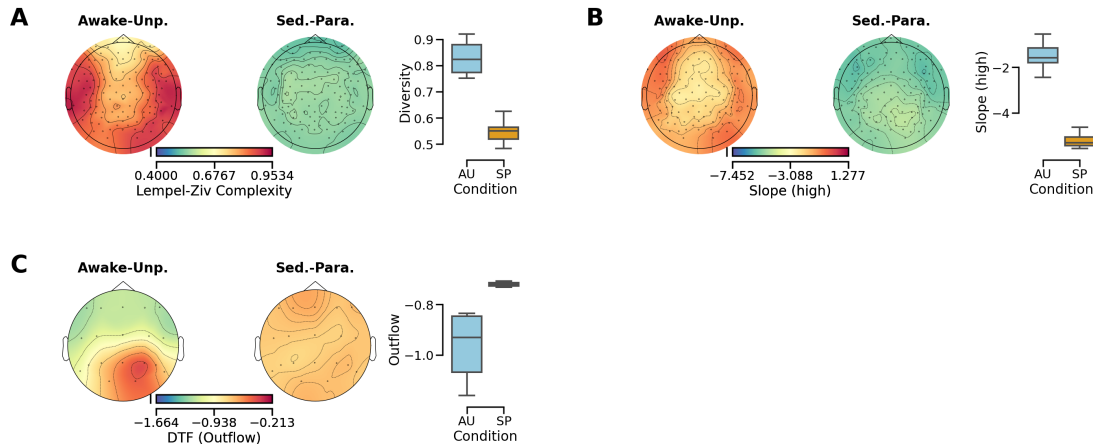
Spontaneous measures



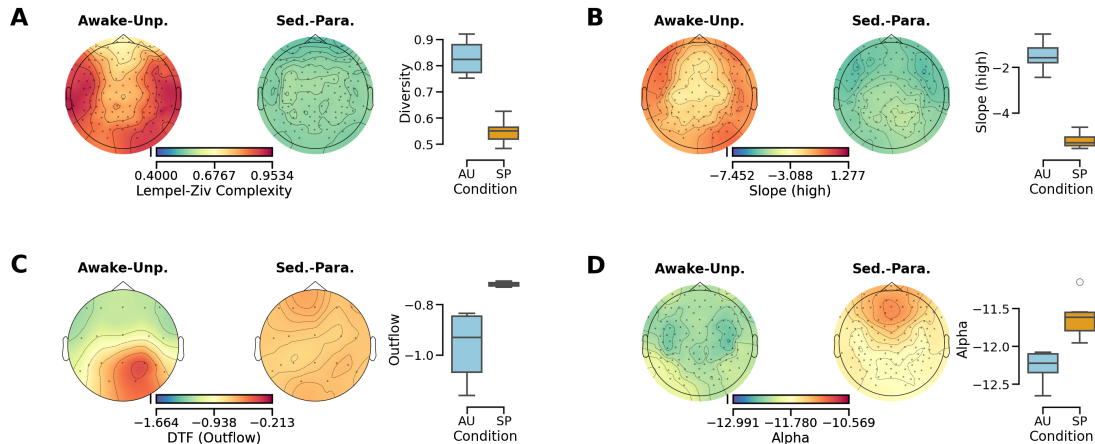
Spontaneous measures



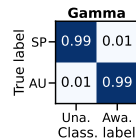
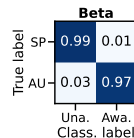
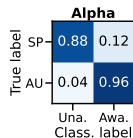
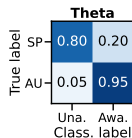
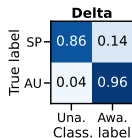
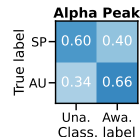
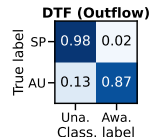
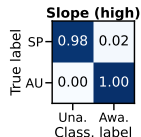
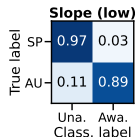
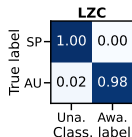
Spontaneous measures



Spontaneous measures



Leave-one-subject out classification



Does this mean we can detect AAGA?

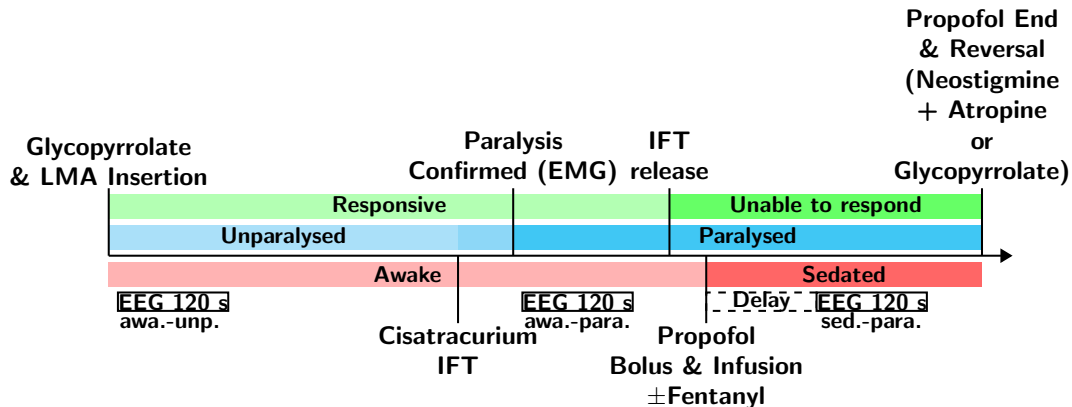
The issue of neuromuscular blockade

- ▶ Neuromuscular blocking agents (NMBAs) are routinely used to improve the quality of surgical conditions¹⁴
- ▶ NMBAs relax the jaw and vocal cord muscles, making it easier to insert the endotracheal tube and reduce the risk of injury during intubation
- ▶ Relaxed muscles, e.g. in the abdomen, make it easier to manipulate and access organs and prevent movement which is essential for delicate surgeries
- ▶ Thus to evaluate if a measure of consciousness is suitable to detect awareness during anaesthesia it must be evaluated with only NMBAs
- ▶ We used a data set recorded by a group recorded by a group in in Australia that investigated the effect of paralysis on the EEG¹⁵

¹⁴Martini, C. H., Boon, M., Bevers, R. F., Aarts, L. P. & Dahan, A. Evaluation of surgical conditions during laparoscopic surgery in patients with moderate vs deep neuromuscular block. BJA: British Journal of Anaesthesia 112, 498–505. doi:10.1093/bja/aet377 (Mar. 2014).

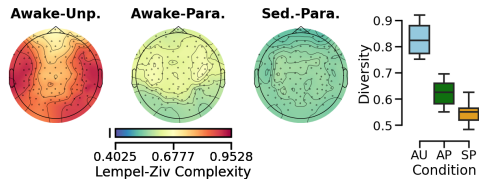
¹⁵Whitham, E. M. et al. Scalp electrical recording during paralysis: Quantitative evidence that EEG frequencies above 20 Hz are contaminated by EMG. Clinical Neurophysiology 118, 1877–1888. doi:10.1016/j.clinph.2007.04.027 (Aug. 2007).

Experimental design

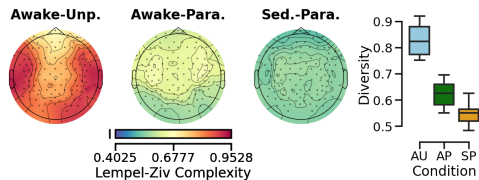
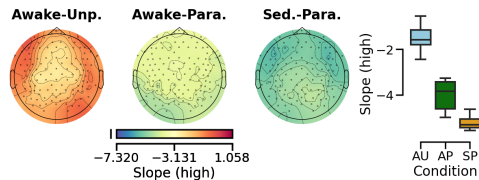


The effect of NMBAs on spontaneous measures

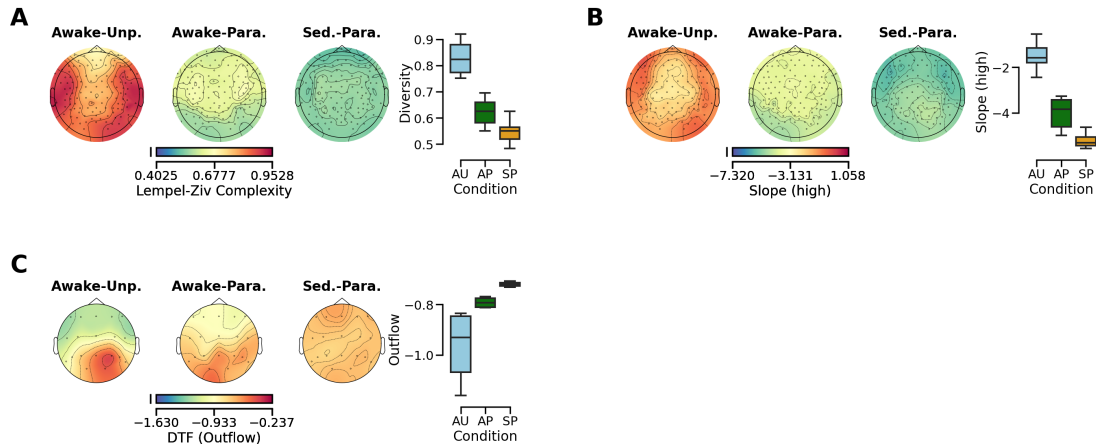
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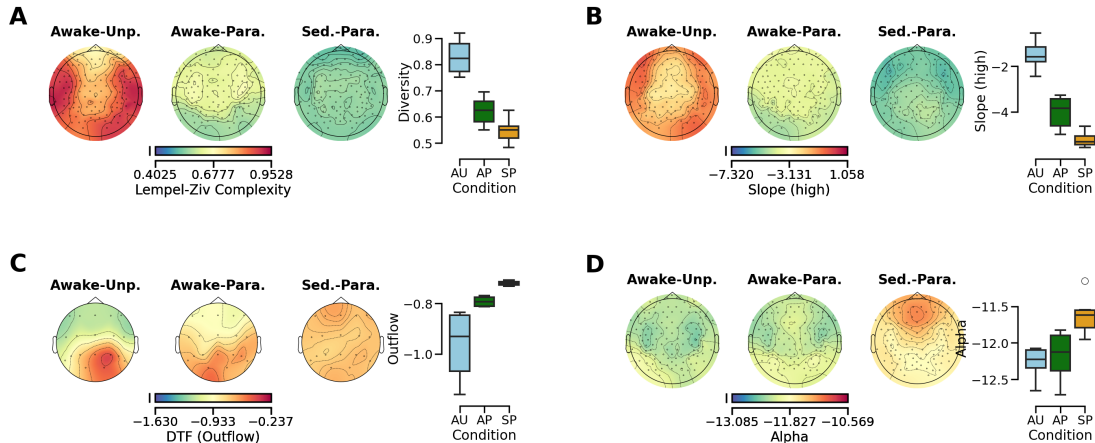
The effect of NMBA on spontaneous measures

A

B


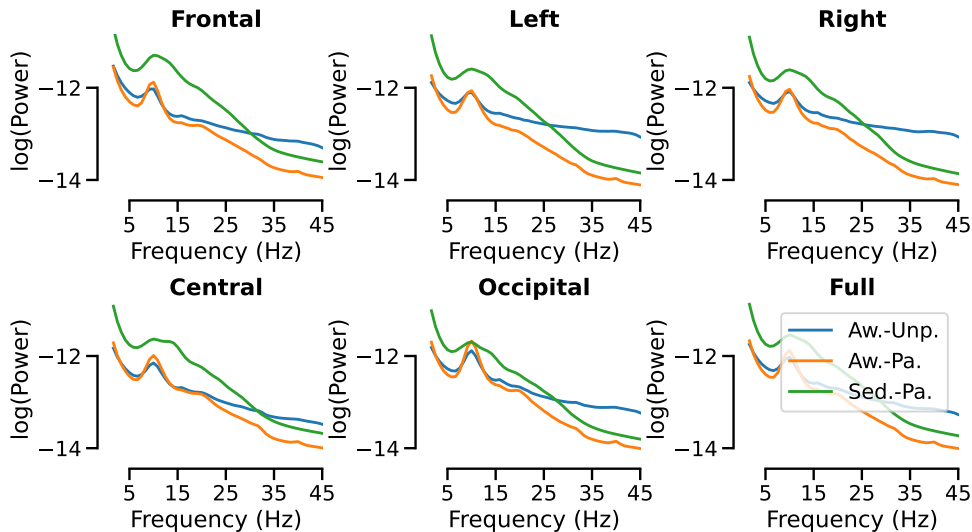
The effect of NMBA's on spontaneous measures



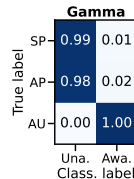
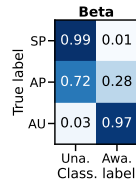
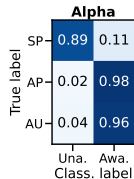
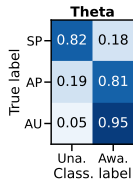
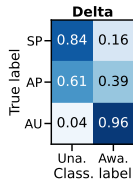
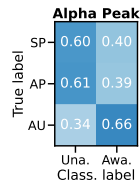
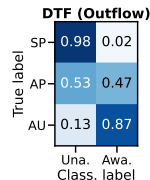
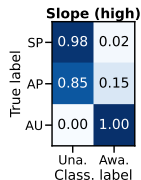
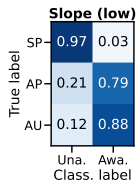
The effect of NMBA on spontaneous measures



The effect of NMBA on PSD



Leave-one-subject out classification including experimental AAGA



Is there nothing we can do?

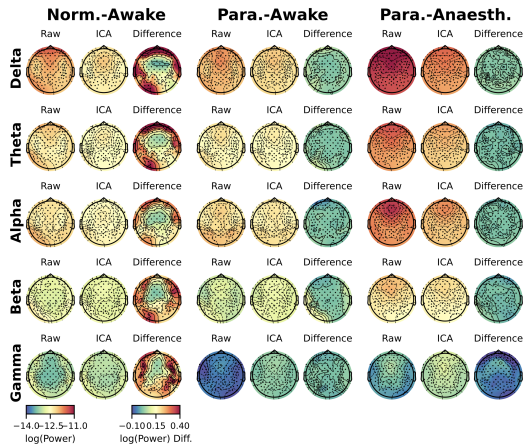
Removing EMG

- ▶ Hypothesis: removal of EMG contamination in 'Normal-Awake' state will improve recognition of 'Paralysed-Awake' as conscious
- ▶ Did not want to overfit a method to EMG components from this dataset
- ▶ Extracted 100 components with extended Infomax ICA
- ▶ Automatic rejection with ICLabel¹⁶
- ▶ Removed everything except 'brain'/'other'

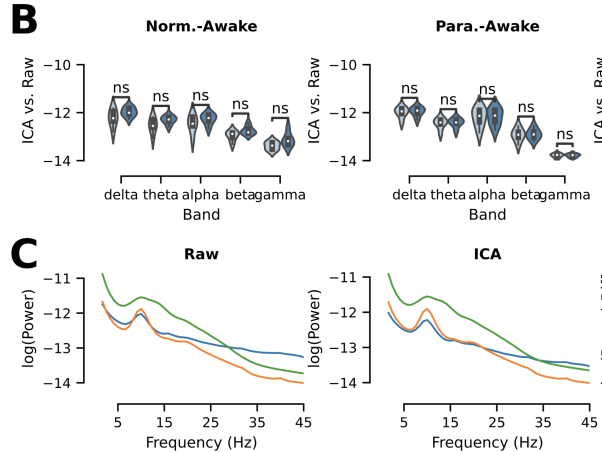
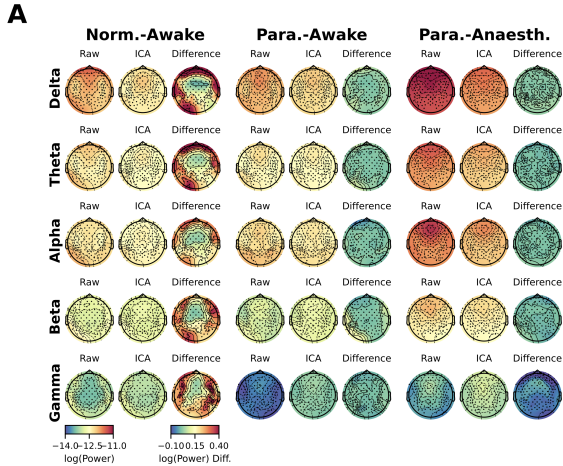
¹⁶Pion-Tonachini, L., Kreutz-Delgado, K. & Makeig, S. ICLabel: An automated electroencephalographic independent component classifier, dataset, and website. *NeuroImage* 198, 181–197. doi:10.1016/j.neuroimage.2019.05.026 (Sept. 2019).

Effect of ICA removal on PSD

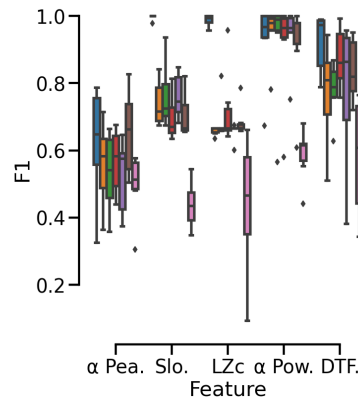
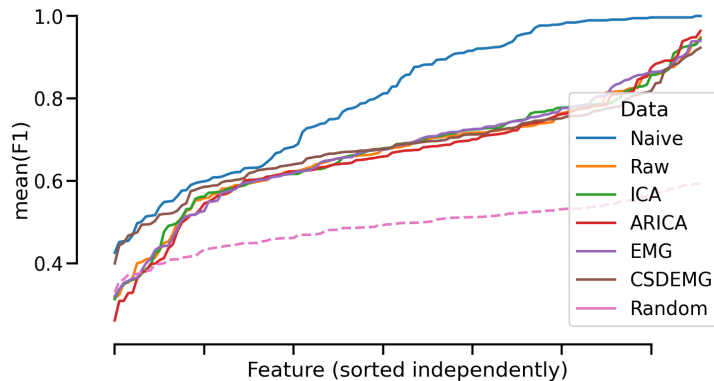
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Effect of ICA removal on PSD

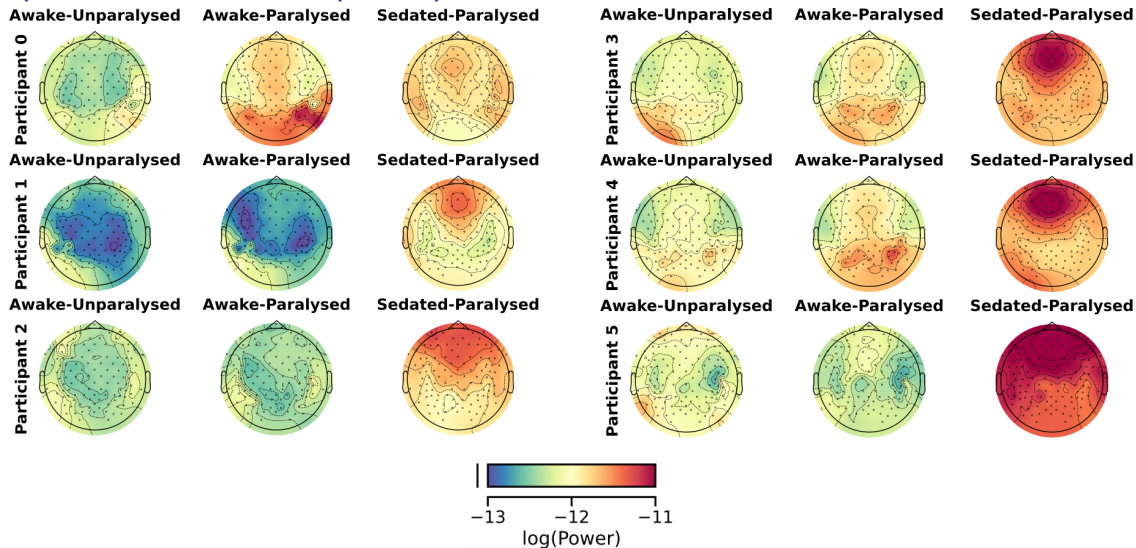


Effect of artifact removal on feature exploration



Is alpha OK then?

Alpha from individual participants



The changes in alpha depend on the anaesthetic

- ▶ Increase of frontal alpha is commonly observed in propofol induced anaesthesia¹⁷
- ▶ This is referred to as alpha anteriorization
- ▶ Alpha power may also decrease:
 - ▶ In about 31% of patients undergoing surgery alpha power drops were observed
 - ▶ Higher doses of propofol may eventually lead to decreasing alpha
- ▶ This may also be drug dependent:
 - ▶ In a study of patients anaesthetised with sevoflurane or desflurane 4% did not show any alpha
 - ▶ Half of the patients showed alpha increase, the other half decreases
 - ▶ The authors suggest using the peak frequency instead of power¹⁸

¹⁷Gutierrez, R., Maldonado, F., Egana, J. I. & Penna, A. Electroencephalographic Alpha and Delta Oscillation Dynamics in Response to Increasing Doses of Propofol. en-US. Journal of Neurosurgical Anesthesiology 34, 79. doi:10.1097/ANA.0000000000000733 (Jan. 2022).

¹⁸Hight, D., Voss, L. J., Garcia, P. S. & Sleight, J. Changes in Alpha Frequency and Power of the Electroencephalogram during Volatile-Based General Anesthesia. Frontiers in Systems Neuroscience 11, 36. doi:10.3389/fnsys.2017.00036 (May 2017).

Conclusion

- ▶ Alpha power is well known to be affected by general anaesthesia
- ▶ Limitations known from other studies apply (see previous slide)
- ▶ Alpha power (especially when limited to frontal electrodes) was the most robust measure (in the sense of recognising the 'Awake-Paralysed' state as aware) in this analysis
- ▶ Nonetheless it is safe to conclude the perfect marker of **environmentally connected consciousness** (using spontaneous EEG) has not yet been found
- ▶ Perturbational methods such as PCI may be an alternative but not for online monitoring and TMS stimulation during surgery may be problematic
- ▶ Can BCIs fill this unmet need? See talk Sébastien!



**Flinders
University**



University of Essex



**UNIVERSITY
OF OSLO**



VESTRE VIKEN

Position

- ▶ Looking for a PostDoc for start in Q3 2026
- ▶ Detection of pain from the EEG recorded during clinical procedures
- ▶ EEG analysis and machine learning (Deep Learning, Riemannian Classifiers) skills required
- ▶ Located in England
- ▶ Travel to collaborators in Norway and for conferences included!
- ▶ 18 Months duration
- ▶ Lots of interesting data!

Thank you for listening!